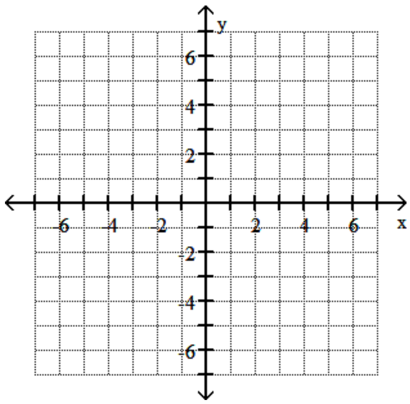
MTH 131 Ashe: Exam 2

Name________________________________

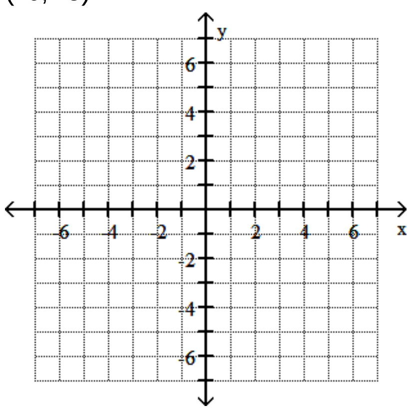
SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Plot the given point in a rectangular coordinate system.

1) (2, -3) 1) _______________

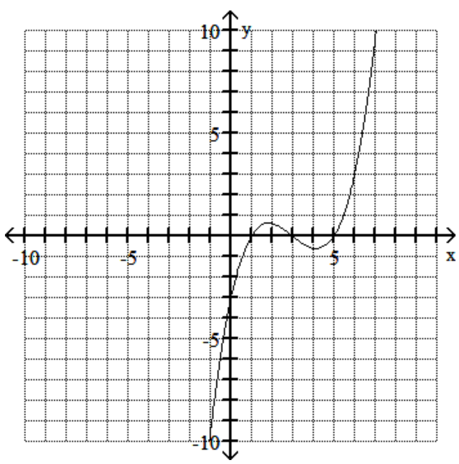



2) (-6, -3) 2) _______________




Use the graph to determine the x- and y-intercepts.

3) 3) _______________



Solve and check the linear equation.

4) $5x - 5 = 30$ 4) ______________

5) $-6x - 10 = -6 + 7x$ 5) ______________

6) $-7x + 5 + 5(x + 1) = -3x + 3$ 6) ______________

Solve the equation.

7) $\frac{x}{6} = \frac{x}{8} + 5$ 7) ______________

8) $\frac{x + 18}{6} + \frac{x + 9}{9} = x - 3$ 8) ______________

First, write the value(s) that make the denominator(s) zero. Then solve the equation.

9) $\frac{7}{x} = \frac{1}{2x} + 26$ 9) ______________

Use the five-step strategy for solving word problems to find the number or numbers described in the following exercise.

10) When 3 times a number is subtracted from 7 times the number, the result is 44. What is the number? 10) ______________

11) When 10% of a number is added to the number, the result is 143. What is the number? 11) ______________

Solve the problem.

12) A car rental agency charges $150 per week plus $0.20 per mile to rent a car. How many miles can you travel in one week for $250? 12) ______________

Solve the formula for the specified variable.

13) $d = rt$ for $r$ 13) ______________

Solve the equation by factoring.

14) $x^2 = x + 42$ 14) ______________

15) $3x^2 - 20x = 7$ 15) ______________

Solve the equation by the square root property.

16) $6x^2 = 54$ 16) ______________

17) $3(x - 9)^2 = 21$ 17) ______________

Solve the equation using the quadratic formula.

18) $x^2 + 5x + 1 = 0$ 18) ____________

19) $4x^2 = -8x - 1$ 19) ____________

Solve the polynomial equation by factoring and then using the zero product principle.

20) $x^3 - 10x^2 + 24x = 0$ 20) ____________

Solve the radical equation, and check all proposed solutions.

21) $\sqrt{x + 1} = 6$ 21) ____________

22) $\sqrt{3x + 28} = x$ 22) ______________

Solve the absolute value equation or indicate that the equation has no solution.

23) $|x - 8| = 4$ 23) ______________

24) $|4x - 2| - 5 = -12$ 24) ______________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Determine whether the equation is an identity, a conditional equation, or an inconsistent equation.

25) $2(2x - 10) = 4x - 20$ 25) ______

A) Identity B) Conditional equation C) Inconsistent equation